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Neo Language Box

Platform overview — prepared for management review

Neo Language Box is a **self-contained Windows desktop application** that listens to a live speaker and produces spoken and written translations in real time, in multiple languages at once. It runs entirely on one laptop — no cloud service, no internet connection, and no external AI subscription required to operate.

RUNTIME **Python, packaged as a native .exe**	PROCESSING **100% on-device**
INTERNET **Not required to run**	PER-USE COST **$0 — no metered API calls**

HOW A SENTENCE MOVES THROUGH THE SYSTEM

01
Capture
Room or system audio

02
Speech → Text
On-device recognition

03
Translate
Local translation engine

04
Text → Speech
Local voice synthesis

05
Deliver
Phones on venue Wi-Fi

WHAT'S ACTUALLY UNDER THE HOOD

APPLICATION	**Python desktop app** — built with PyInstaller and an Inno Setup installer. One .exe; no admin rights required to install.
SPEECH RECOGNITION	**Open-source speech-to-text models** — run locally on the laptop's own CPU, no cloud call involved.
TRANSLATION	**Specialised machine-translation models** — purpose-built for translation, not a general chatbot-style AI. Each language runs its own lightweight model.
VOICE SYNTHESIS	**Offline text-to-speech models** — one trained voice per target language, also generated locally.
DELIVERY TO LISTENERS	**Local Wi-Fi + web server** — listeners open a link in their phone's own browser. Nothing to install on their end.

WHY THIS APPROACH, NOT A CLOUD AI SERVICE

Private by design

Audio and translated text never leave the building — nothing is sent to a third-party server.

No recurring cost

Translation and voice generation aren't metered. One install, no per-minute or per-word billing.

Works offline

No dependency on venue internet quality or a vendor's uptime — it runs the same with no connection at all.

Prepared for an internal management presentation. Current build translates into 12 languages live (plus the original English), with more addable as voice models are validated.